Studio

Electric

Everything the electrician needs to wire one container studio: the first fix before the walls close, the second fix, the tests and the certificate.

Status: revision A2, 2026-10-28, PRELIMINARY. Every rating is design intent until a RURA-licensed electrician (class A) designs, installs and certifies the installation to RS 565-1:2011.

The two studios are wired the same, so one page serves both.

Kinyarwanda: soma aya mabwiriza mu Kinyarwanda · Print: this page as a PDF · The circuits drawn: the studio page · When it happens: the execution plan

At a glance

- **End A**: the closed end wall, by the bed. **End B**: the cargo-door end, by the bathroom.
- **Back wall**: the long wall with no openings. **Front wall**: the long wall with the sliding glass door.
- Positions in the points table are millimetres from the finished end-A wall (x), from the finished back wall (y) and up from the finished floor (z).

What	Design
Supply	one inlet 63 A, IP67, IEC 60309 blue (2P+E), outside the back wall straight behind the board; any 230 V source
Main	63 A double-pole breaker (C curve) and a DIN kWh check meter; tails 16 mm² through a gland
Board	24 modules in two rows of 12: main, check meter, type 2 SPD, priority relay and its contactor, one RCBO (30 mA, type A) per circuit
Circuits	8, C1 to C8 (table below)
Maximum demand	45.3 A diversified: lights 0.3 + fans 0.3 + sockets 16.4 + the larger of the hob 11.6 or the instant shower heater 23.9 with the AC 4.3 (D-053)
Earthing	TT: one earth rod per container; never a combined neutral-earth (PME) supply
Cable route	20 mm conduit in the batten cavities, buried in spray foam after the first fix

When	What	Step
After the battens (S03)	First fix: earth rod, conduits, boxes, cables	S04
After the lining and the kitchen	Second fix: board, devices, tests, certificate	S19

The kWh meter in each studio is a check meter for monitoring only: electricity is in the room rate.

Hold points

1. **Battens first.** Start the first fix only when the battens (S03) are fixed and the floor sheets are down.
2. **Before the foam.** Every conduit, box and cable is in, insulation-tested, photographed wall by wall and signed before the spray foam (S05). The foam buries the conduits: nothing can be added later except by pulling through them.
3. **Before power-on.** The tests pass and the RURA certificate is signed before the inlet is ever fed.

Steps

S04: first fix

1. Set the board box on the back wall at the galley, and the inlet outside straight behind it: positions DB and PI in the points table.
2. Drive the earth rod (E1) outside end B and run its conductor to the board's earth bar; measure its resistance later, at the tests.
3. Run 20 mm conduit up from the board into the ceiling void, along it, then straight down to each box. No diagonal runs; fix every box to a batten.
4. Pull the cables per circuit: 1.5 mm² for C1 and C6, 2.5 mm² for C2, C3, C7 and C8, 4 mm² for C4, 6 mm² for C5 (the instant shower heater).
5. Take C6, the outdoor lights, out through one gland in the solid front strip at the bed end, then in UV-stable conduit outside.
6. Bathroom: an IP65 light and an IPX4 extract fan (zone 1, above the tray and below 2.25 m), no sockets, switched from outside the door.
7. Write each point's id on the steel beside its box, so the carpenter cuts the lining panels in the right places.
8. Test continuity and insulation resistance on every cable, photograph each wall, and only then sign the release for the foam.

S19: second fix

1. Fit the board: the main breaker and check meter, the SPD, the priority relay and contactor, and the RCBOs (B6 for C1 and C6, B16 for C2, C3, C7 and C8, B20 for C4, B25 for C5).
2. Fit the devices: lights, fans, sockets, the hob outlet (P9, isolated at its RCBO on the board, in sight), the smoke and heat alarms (AL1, AL2).
3. Bond the container body, the front deck, the stair and the roof steelwork to the main earthing terminal.
4. Check the priority relay: with the hob on, the shower heater (C5) and the AC (C7) must drop out. The contactor is 40 A: it carries the heater's 23.9 A.

Tests before power-on

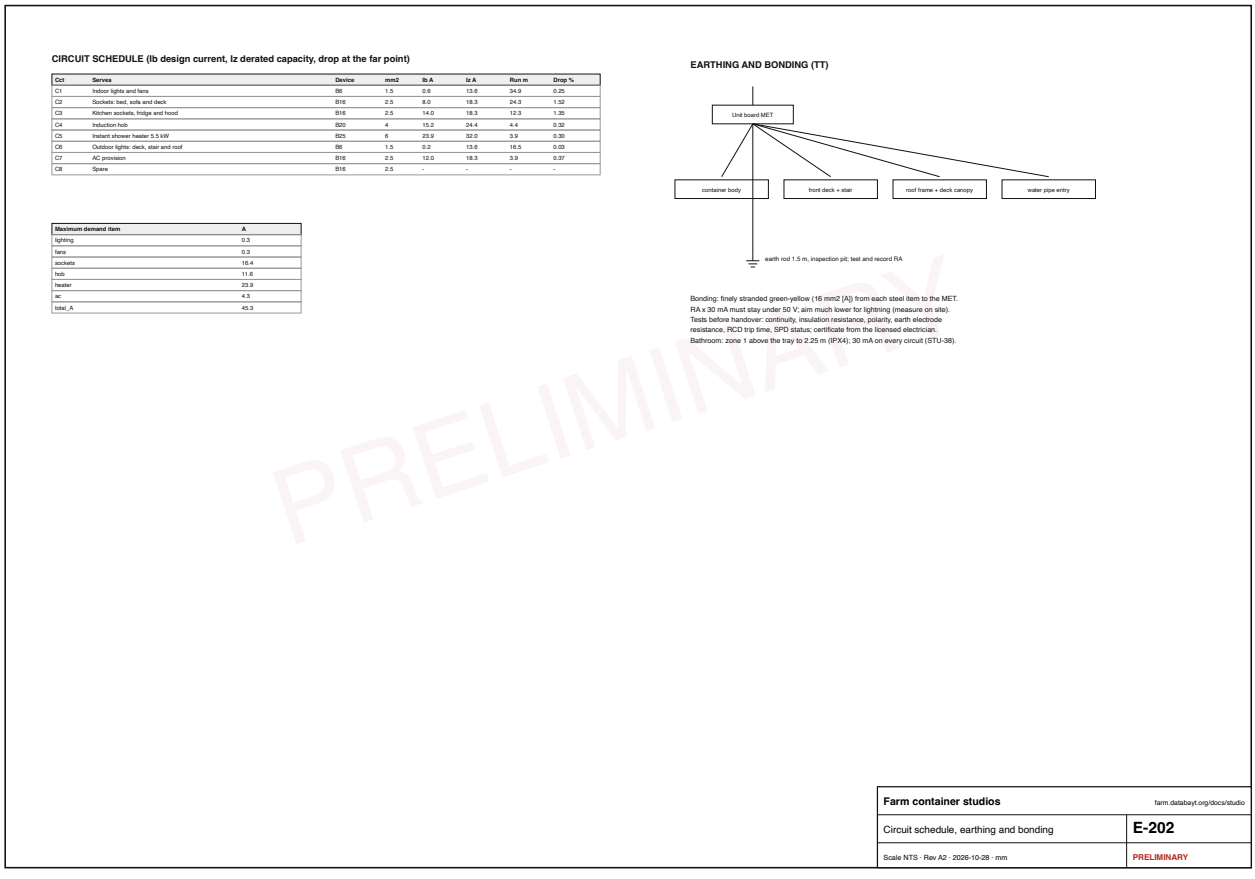
1. Continuity of every protective conductor.
2. Insulation resistance of every circuit.
3. Polarity at every socket and light.
4. Earth-electrode resistance of E1: write the value down.
5. The trip time of every RCBO.
6. The SPD status window, and the priority relay once more.
7. The RURA certificate, before the inlet is fed.

Circuits

Each circuit is checked by the model: its current fits the breaker, the breaker fits the cable, and the voltage drop stays under 3 % for lights and 5 % for the rest.

Circuit	Serves	Breaker	Cable	Design current	Run	Voltage drop	Points
C1	Indoor lights and fans	B6	1.5 mm²	0.6 A	34.9 m	0.25 %	L1 L2 L3 L4 L5 L7 S1 S2 S3 F1 F2
C2	Sockets: bed, sofa and deck	B16	2.5 mm²	8 A	24.3 m	1.52 %	P1 P2 P3 P5

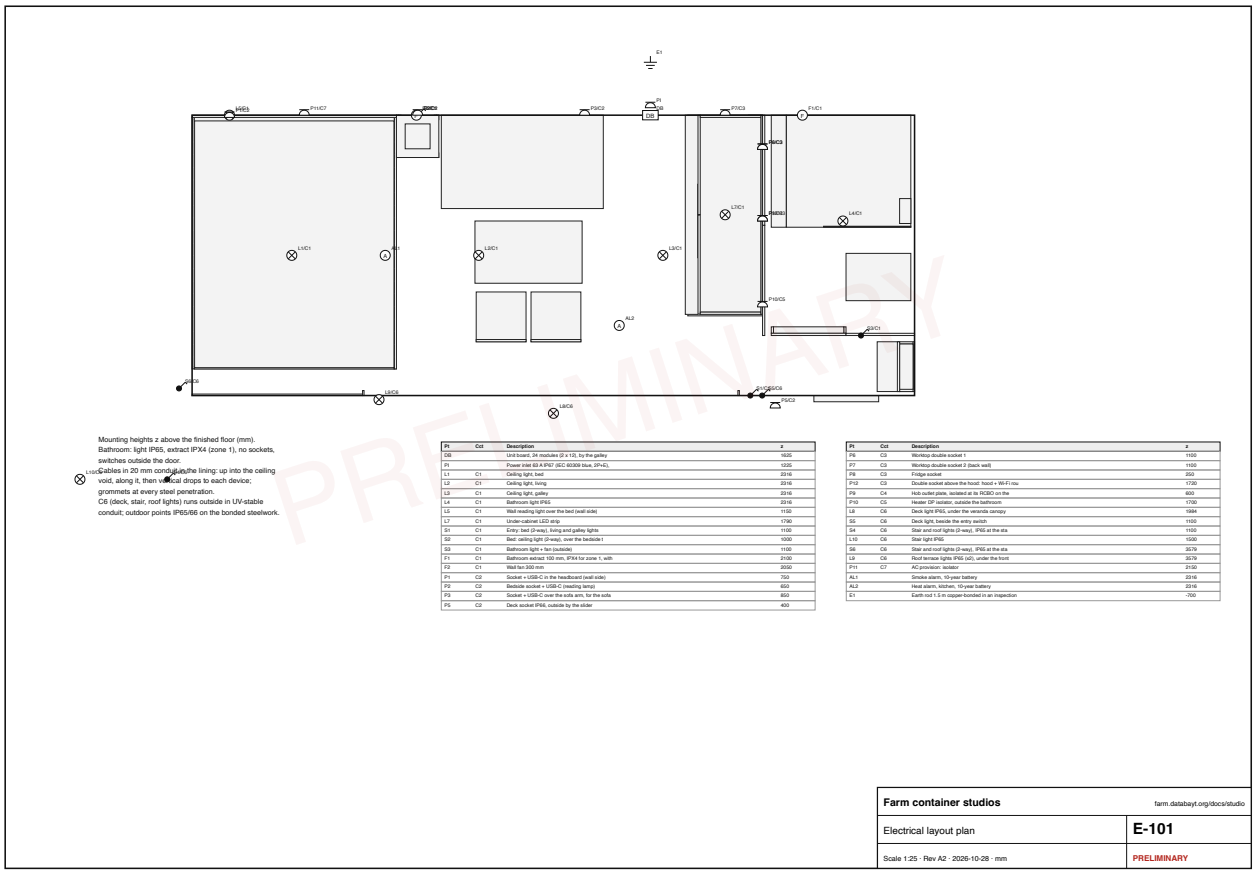
Circuit	Serves	Breaker	Cable	Design current	Run	Voltage drop	Points
C3	Kitchen sockets, fridge and hood	B16	2.5 mm²	14 A	12.3 m	1.35 %	P6 P7 P8 P12
C4	Induction hob	B20	4 mm²	15.2 A	4.4 m	0.32 %	P9
C5	Instant shower heater 5.5 kW	B25	6 mm²	23.9 A	3.9 m	0.3 %	P10
C6	Outdoor lights: deck, stair and roof	B6	1.5 mm²	0.2 A	16.5 m	0.03 %	L8 S5 S4 L10 S6 L9
C7	AC provision	B16	2.5 mm²	12 A	3.9 m	0.37 %	P11
C8	Spare	B16	2.5 mm²	–	–	–	spare



E-202 · Circuit schedule, earthing and bonding · open full size

Points

33 points on the plan. The bathroom has an IP65 light, an IPX4 extract fan and no sockets, switched from outside the door.



E-101 · Electrical layout plan · open full size

Point	What	Type	Circuit	Height	Load
DB	Unit board, 24 modules (2 x 12), by the galley	board	–	1,625 mm	–
PI	Power inlet 63 A IP67 (IEC 60309 blue, 2P+E), outside behind the board	power inlet	–	1,225 mm	–
L1	Ceiling light, bed	light	C1	2,316 mm	12 W
L2	Ceiling light, living	light	C1	2,316 mm	12 W
L3	Ceiling light, galley	light	C1	2,316 mm	12 W
L4	Bathroom light IP65	light	C1	2,316 mm	12 W
L5	Wall reading light over the bed (wall side)	light	C1	1,150 mm	4 W

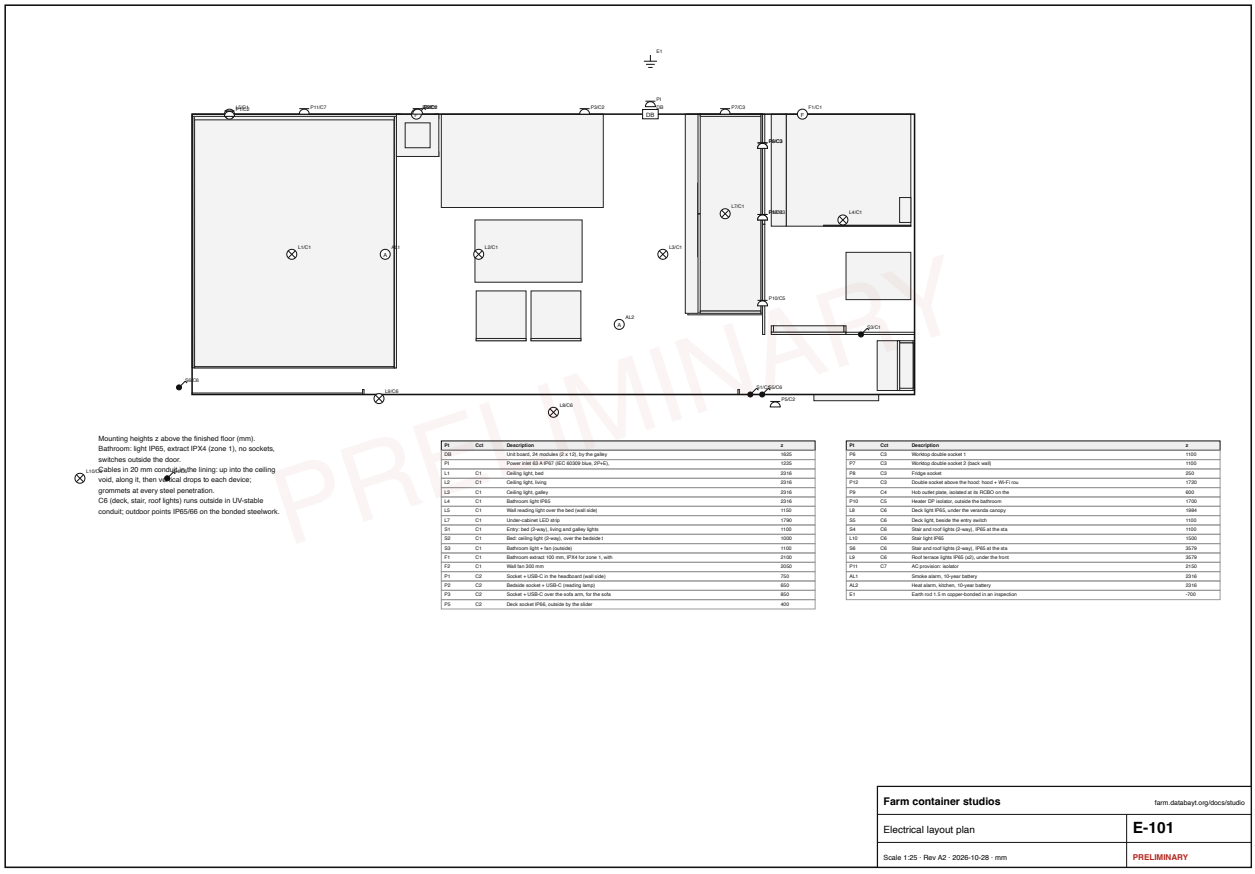
Point	What	Type	Circuit	Height	Load
L7	Under-cabinet LED strip	light	C1	1,790 mm	10 W
S1	Entry: bed (2-way), living and galley lights	switch	C1	1,100 mm	–
S2	Bed: ceiling light (2-way), over the bedside table	switch	C1	1,000 mm	–
S3	Bathroom light + fan (outside)	switch	C1	1,100 mm	–
F1	Bathroom extract 100 mm, IPX4 for zone 1, with run-on	fan	C1	2,100 mm	20 W
F2	Wall fan 300 mm	fan	C1	2,050 mm	50 W
P1	Socket + USB-C in the headboard (wall side)	socket + USB-C	C2	750 mm	200 W
P2	Bedside socket + USB-C (reading lamp)	socket + USB-C	C2	650 mm	200 W
P3	Socket + USB-C over the sofa arm, for the sofa and the table	socket + USB-C	C2	850 mm	500 W
P5	Deck socket IP66, outside by the slider	socket	C2	400 mm	300 W
P6	Worktop double socket 1	socket	C3	1,100 mm	2,000 W
P7	Worktop double socket 2 (back wall)	socket	C3	1,100 mm	1,000 W
P8	Fridge socket	socket	C3	250 mm	100 W
P12	Double socket above the hood: hood + Wi-Fi router	socket	C3	1,720 mm	120 W
P9	Hob outlet plate, isolated at its RCBO on the board	outlet	C4	600 mm	3,500 W
P10	Heater DP isolator, outside the bathroom	outlet	C5	1,700 mm	5,500 W
L8	Deck light IP65, under the veranda canopy	light	C6	1,984 mm	10 W
S5	Deck light, beside the entry switch	switch	C6	1,100 mm	–
S4	Stair and roof lights (2-way), IP65 at the stair foot	switch	C6	1,100 mm	–
L10	Stair light IP65	light	C6	1,500 mm	8 W

Point	What	Type	Circuit	Height	Load
S6	Stair and roof lights (2-way), IP65 at the stair head	switch	C6	3,579 mm	–
L9	Roof terrace lights IP65 (x2), under the front top rail	light	C6	3,579 mm	20 W
P11	AC provision: isolator	outlet	C7	2,150 mm	1,000 W
AL1	Smoke alarm, 10-year battery	alarm	–	2,316 mm	–
AL2	Heat alarm, kitchen, 10-year battery	alarm	–	2,316 mm	–
E1	Earth rod 1.5 m copper-bonded in an inspection pit, behind the board	earth rod	–	-700 mm	–

Open questions

1. Which source feeds the inlet: the site board, a generator or a solar inverter? Any 230 V, 50 Hz source can, if it protects its own cable at 63 A or less; the farm's **electric** page decides.

Drawings



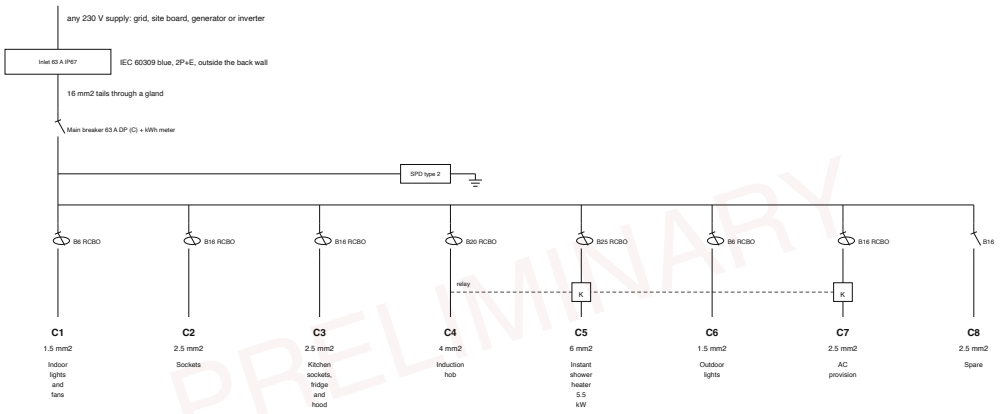
Mounting heights z above the finished floor (mm).
Bathroom: light IP65, extract IPX4 (zone 1), no sockets, switches outside the door.
Cables in 20 mm conduit in the lining: up into the ceiling void, along it, then vertical drops to each device; grommets at every steel penetration.
C6 (deck, stair, roof lights) runs outside in UV-stable conduit; outdoor points IP65/66 on the bonded steelwork.

Pt	Cct	Description	z
DB		Unit board, 24 modules (2 x 12), by the galley	1625
P1		Power inlet 63 A IP67 (IEC 60309 blue, 3P+E),	1235
L1	C1	Ceiling light, bed	2316
L2	C1	Ceiling light, living	2316
L3	C1	Ceiling light, galley	2316
L4	C1	Bathroom light IP65	2316
L5	C1	Wall reading light over the bed (wall side)	1150
L7	C1	Under-cabinet LED strip	1790
S1	C1	Entry / bed (2-way), living and galley lights	1100
S2	C1	Bed: ceiling light (2-way), over the bedside t	1030
S3	C1	Bathroom light + fan (outside)	1100
F1	C1	Bathroom extract 100 mm, IPX4 for zone 1, with	2100
F2	C1	Wall fan 300 mm	2050
P1	C2	Socket + USB-C in the headboard (wall side)	750
P2	C2	Bedside socket + USB-C (reading lamp)	650
P3	C2	Socket + USB-C over the sofa arm, for the sofa	850
P5	C2	Deck socket IP66, outside by the slider	400

Pt	Cct	Description	z
P6	C3	Worktop double socket 1	1100
P7	C3	Worktop double socket 2 (back wall)	1100
P8	C3	Fridge socket	250
P12	C3	Double socket above the hood: hood + Wi-Fi rou	1720
P9	C4	Hob outlet plate, isolated at its RCBO on the	600
P10	C5	Heater DP isolator, outside the bathroom	1700
L8	C6	Deck light IP65, under the veranda canopy	1984
S5	C6	Deck light, beside the entry switch	1100
S4	C6	Stair and roof lights (2-way), IP65 at the sta	1100
L10	C6	Stair light IP65	1500
S6	C6	Stair and roof lights (2-way), IP65 at the sta	3579
L9	C6	Roof terrace lights IP65 (x2), under the front	3579
P11	C7	AC-provision: isolator	2150
AL1		Smoke alarm, 10-year battery	2316
AL2		Heat alarm, kitchen, 10-year battery	2316
E1		Earth rod 1.5 m copper-bonded in an inspection	-700

E-101 Electrical layout plan · open full size

ONE STUDIO, FROM ITS INLET TO EVERY CIRCUIT



PRELIMINARY

The studio plugs in: any 230 V source protected at 63 A or less; the 63 A main breaker protects the inlet and tails whatever feeds them.
Priority relay on C4: while the hob runs, contactor K sheds C5 and C7 (heater and AC).
Diversified maximum demand about 45 A (AS-19), within the 63 A inlet. One RCBO 30 mA type A per circuit.
TT earthing: the studio's own earth rod at the board; never rely on the supply's earth (STU-37).
Ratings are design intent: a RURA class A electrician designs and certifies (STU-15), to RS 565-1:2011 (STU-14).

Farm container studios	farm.databayt.org/docs/studio
Single-line diagram: inlet and unit board	**E-201**
Scale NTS · Rev A2 · 2026-10-28 · mm	**PRELIMINARY**

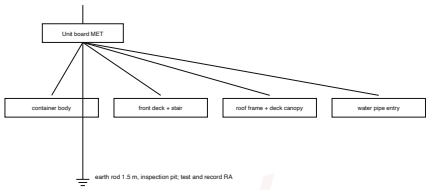
E-201 Single-line diagram: inlet and unit board · open full size

CIRCUIT SCHEDULE (Ib design current, Iz derated capacity, drop at the far point)

Cct	Serves	Device	mm2	Ib A	Iz A	Run m	Drop %
C1	Indoor lights and fans	B6	1.5	0.6	13.6	34.9	0.25
C2	Sockets: bed, sofa and deck	B16	2.5	8.0	18.3	24.3	1.52
C3	Kitchen sockets, fridge and hood	B16	2.5	14.0	18.3	12.3	1.35
C4	Induction hob	B20	4	15.2	24.4	4.4	0.32
C5	Instant shower heater 5.5 kW	B25	6	23.9	32.0	3.9	0.30
C6	Outdoor lights: deck, stair and roof	B6	1.5	0.2	13.6	16.5	0.03
C7	AC provision	B16	2.5	12.0	18.3	3.9	0.37
C8	Spare	B16	2.5	-	-	-	-

Maximum demand item	A
lighting	0.3
fans	0.3
sockets	16.4
hob	11.6
heater	23.9
ac	4.3
total_A	45.3

EARTHING AND BONDING (TT)



earth rod 1.5 m, inspection pit; test and record RA

Bonding: finely stranded green-yellow (16 mm2 [Al]) from each steel item to the MET. RA x 30 mA must stay under 50 V; aim much lower for lightning (measure on site). Tests before handover: continuity, insulation resistance, polarity, earth electrode resistance, RCD trip time, SPD status; certificate from the licensed electrician. Bathroom: zone 1 above the tray to 2.25 m (IPX4); 30 mA on every circuit (STU-38).

PRELIMINARY

Farm container studios	farm.databayt.org/docs/studio
Circuit schedule, earthing and bonding	E-202
Scale NTS · Rev A2 · 2026-10-28 · mm	PRELIMINARY

E-202 Circuit schedule, earthing and bonding · open full size